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Charles E. Kahn, Jr., MD, MS
Editor, Radiology: Artificial Intelligence
Radiological Society of North America

Dear Dr Kahn,

Title. “Supervised Tumor Attention Transfers Across CT Cohorts, but EGFR Prediction Does Not Exceed Clinical Variables.”

Authors. [Complete author list, in final order, with degrees.]

Article type. Original Research.

We submit this manuscript for your consideration. The CT radiogenomics literature reports EGFR prediction at AUC 0.80–0.89, almost always from single-center cohorts without external validation. We took an image encoder whose attention is supervised against tumor segmentations and separated two claims this literature routinely bundles: that such a model attends to the lesion, and that it genotypes it. The attention transferred to 420 patients in another country, on another scanner fleet, in another treatment setting. Genotype prediction did not: pooled AUC was 0.618 against 0.794 for smoking status alone, adding the model to a five-variable clinical model changed AUC by -0.029 , no fusion topology helped, and calibration was worse than predicting the base rate.

We think the useful result is the dissociation rather than the null. A saliency map that tracks the tumor on an unseen institution’s data, attached to a classifier performing at 0.618, is evidence that plausible localization is not evidence of predictive validity — and that can only be observed when the prediction fails, so it cannot appear in studies that report both as successes.

Two objections we would rather raise than have noticed. First, the internal cohort is 153 patients with a known EGFR call and the pre-specified primary analysis set is 97, far below what the journal typically carries. Our argument is not that this is adequate but that the question is not resolvable by enlarging it: Riley’s criterion requires 414 patients for the five-variable clinical model alone, the entire open world of EGFR-labeled CT gives a 95% confidence-interval half-width of 0.154, and the largest external study yet published (1646 patients, 11 473 lesions, Eur Radiol 2026) reproduces our interval at ten times the sample size with our internal cohort as its external one. Second, the primary endpoint clears its pre-specified threshold by a single fold (11 of 15). Table 3 therefore reports every evaluation individually; the distribution is bimodal rather than marginal, with the weakest success at 0.757 pointing accuracy and the strongest failure at exactly 0.000.

Two things a reviewer would ask for, done in advance. The crops are lesion-centered, so a center prior is a serious control rather than a courtesy. We therefore translated each external crop and its reference contour so the tumor was no longer centered: across the aligning folds pointing accuracy fell from 0.831 to 0.711 at a 35% displacement, while the center prior fell from 0.476 to 0.002.

Separately, because 15 fold-checkpoints on one external cohort are not 15 independent studies, we reran the endpoint as a patient-level bootstrap. It favours the model when the training runs are held fixed and does not survive treating the run as a random effect. We report that against our own interest, and it is the paper's main limitation.

Subject overlap with prior work (required). Both imaging collections are public and have been analyzed by other groups, including in each collection's source publication and in the multi-tumor external-validation study cited above, in which NSCLC-Radiogenomics serves as the external cohort. No patient in this analysis overlaps with any published, in-press, or under-review work by these authors. What is new here is the pre-specified separation of the localization and genotyping claims, the three-control external localization evaluation, and the measured precision bound. This overlap is also reported in Materials and Methods. [If any overlapping article exists, upload its PDF at submission and describe here what is new and different.]

Conflicts of interest and industry support (required). [REQUIRED — state all conflicts and any industry support, or state that there are none. ICMJE disclosure forms are submitted for every author.]

Sole submission (required). This manuscript is original, has not been published in the same or similar form by the authors, and has not been and will not be submitted to another journal by the authors or by colleagues at their institutions while it is under consideration at Radiology: Artificial Intelligence. [State here if any similar material has been provided to news or advertising media, and supply a copy.]

Use of artificial intelligence in the study and in manuscript preparation (required by RSNA Editorial Policies). Generative AI was used substantially in this work, and we would rather state the extent than have it inferred.

Tool. Claude Opus 5, accessed through the Claude Code command-line interface (Anthropic, San Francisco, Calif). Dates of access. July 20 to August 13, 2026. Use. Writing and refactoring the analysis code; executing and summarising the reported analyses; literature search; drafting and revising the manuscript, the figures' generating code, and the reporting checklist. Sixty-five of the 68 commits in the repository record that assistance in their commit metadata, so the claim is auditable rather than asserted.

What the authors did. The authors defined the research question, approved the pre-specified analysis plan before external evaluation, verified every reported number against the archived artifact from which it is regenerated, and take full responsibility for the integrity and accuracy of the work. No AI tool meets ICMJE authorship criteria and none is listed as an author.

Errors the assistance introduced. We think a disclosure that reports no errors is describing its review process rather than its work, so: the assistance applied a published sample-size formula incorrectly and the wrong minimum- n reached a draft; it mis-attributed a cited paper, describing a blood-brain-barrier reconstruction study as a mutation-prediction study; and a stale command-line default caused the fusion comparison to be computed on a variant model rather than the reported one, producing an internally inconsistent table. Each was caught before submission, each is recorded with its correction in the public version history, and each has a regression test written in response. The last of these was found by an adversarial review of the manuscript, which is also disclosed here because it materially improved the paper.

Control of the data and the analyses (required by RSNA Editorial Policies). [REQUIRED — name

the author(s) who had full access to and control of the data and who performed the statistical analyses, e.g. “[Initials] had full access to all study data and takes responsibility for the integrity of the data and the accuracy of the analysis.”]

Dual first authorship. [Delete if not applicable. Otherwise state which two authors contributed equally and why equal contribution is warranted; this must also appear on the full title page.]

Prior presentation and preprint posting. [REQUIRED disclosure at submission — state any presentation at a scientific meeting and any preprint posting with server, DOI and date, or state that there has been none.]

Reproducibility, for the editorial office. Every number in the manuscript is regenerated from archived out-of-fold predictions by a single script, requiring no imaging data and no GPU. The repository holds the pre-specified analysis plan with its deviation log — including the deviations that went against us — the completed CLAIM checklist, and all archived runs, negative results included. Because review is double-anonymized, the manuscript cites an anonymized mirror; the permanent URL is [REAL REPOSITORY URL] and the archival DOI is [ZENODO DOI], given here for the editorial office only.

Image licensing. NSCLC-Radiogenomics is licensed CC BY 3.0, and its images appear in Figure 6 with attribution. NSCLC-Radiomics is licensed CC BY-NC 3.0, so no pixel data from it appears in any figure — only derived statistics.

Ethics. [REQUIRED — insert the institutional review board’s written determination. The analysis used only publicly available, fully de-identified data distributed by The Cancer Imaging Archive under open licenses, and no protected health information was accessed.]

Yours sincerely,

[Corresponding author name, degrees]
on behalf of all authors